WASP-35b

R-1708

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_WASP-35_150101` · created 2026-08-02T15:08:55+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457023.74038 +/- 0.00018 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.165 +/- 0.048
Transit depth (Rp/Rs)^2	2.71 +/- 1.57 [%]
Orbital Inclination (inc)	86.26 +/- 2.01
Airmass coefficient 1 (a1)	2.71 +/- 1.81
Airmass coefficient 2 (a2)	-0.77 +/- 0.67
Scatter in the residuals of the lightcurve fit is	63.89 %
Best Comparison Star	None
Optimal Aperture	2.68
Optimal Annulus	4.16
Transit Duration (day)	0.119 +/- 0.025

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.165 ± 0.048 vs 0.1241 (deviation 0.73σ)
Duration fit vs published	0.119 d vs 0.1317 d (deviation 0.43σ)
Mid-time O–C	0.4 min
Depth SNR	2.9
Residual scatter	63.89 %

Light curve

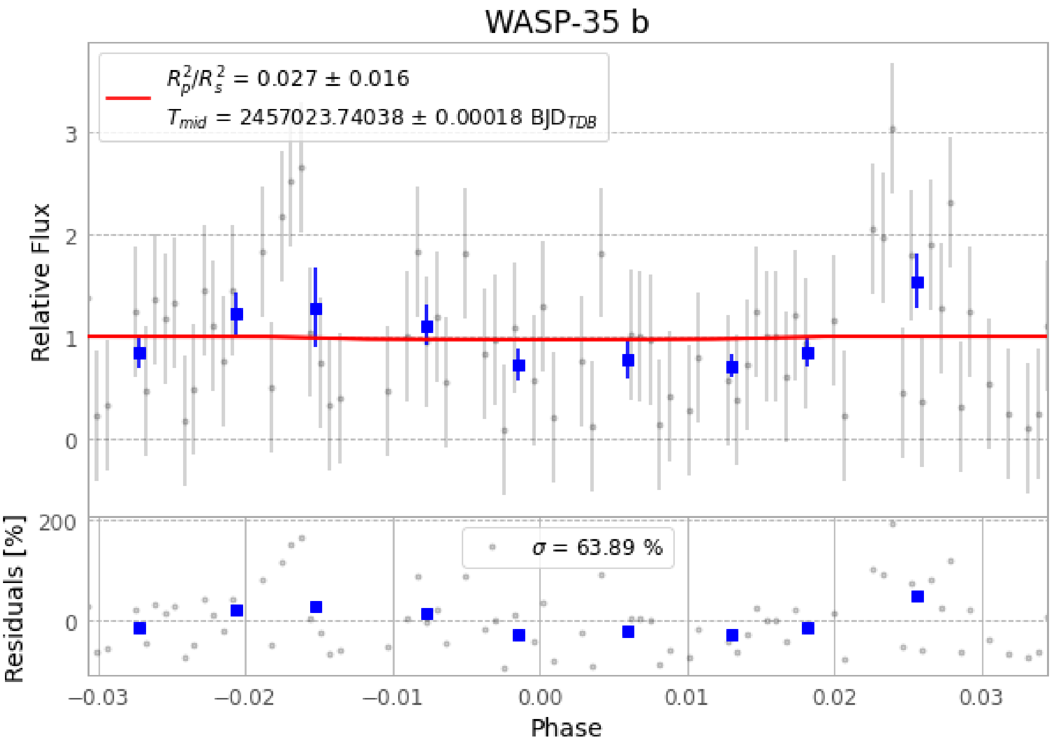

Figure 1 — FinalLightCurve_WASP-35 b_2014-12-31.png (SHA-256 418a1b7f55610de7…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [415, 110], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 c51443bcaec8cebb…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b`

Data provenance

102 FITS frames (67 MB), `WASP-35150101031812.FITS` → `WASP-35150101082112.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-35-150101.log (f4da594b2a02…), FinalLightCurve_WASP-35 b_2014-12-31.png (418a1b7f5561…), FinalLightCurve_WASP-35 b_2014-12-31.csv (3af2af9c1f25…)

Compute resources

Wall time 200.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)